

CAREER PATHS

Qualified structural engineers often specialize in working on one type of building or material—oil platforms or concrete structures, for example. With experience, many move into managing construction projects or become consultants.

INTERN You can study for an undergraduate engineering degree on the job while working as an intern, or in a co-op program.



GRADUATE You will need an accredited degree in a subject such as civil or structural engineering to apply for jobs. Graduate training programs are available.



STRUCTURAL ENGINEER

After gaining experience in junior roles and passing your professional exams, you can practice as a qualified structural engineer. You can then choose to design a variety of buildings or bridges, or select a specialized area to work in.



PROJECT MANAGER

Liaises closely with all of the personnel on a construction project, ensuring that everything happens on time and to brief. A project manager may work independently or as leader of a team.



CONSERVATION AND RESTORATION ENGINEER

Works on the conservation and restoration of historic buildings and structures, combining old and new construction methods.



HUMANITARIAN ENGINEER

Contributes to disaster relief work in the reconstruction of infrastructure and buildings damaged by natural disasters.



FORENSIC ENGINEER

Investigates the reasons for failure or collapse of a structure in situations such as criminal damage, human error, or natural disaster.



SEISMIC ENGINEER

Designs buildings in earthquake-prone countries to ensure that they can cope with seismic movements in the ground in order to minimize building damage and improve safety.

SKILLS GUIDE



Good communication skills—both verbal and written—to deal with clients and prepare reports.



The ability to use mathematical analysis to determine whether a structure can withstand loads.



Problem-solving abilities to spot issues and propose and design solutions throughout project.



Budgeting expertise and commercial awareness of business implications of design decisions.



Excellent organizational skills to schedule and fulfil all stages of the planning and design process.